

Screenshots:

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1. Total data rows: 34
2a. Valid rows: 30
2b. Invalid total: 4

3. Visit count by city:
defaultdict(<class 'int'>, {'New York': 1, 'Los Angeles': 1, 'Chicago': 1, 'Houston': 1, 'Phoenix': 1, 'Philadelphia': 1, 'San Antonio': 1, 'San Diego': 1, 'Dallas': 1, 'San Jose': 1, 'Austin': 1, 'Jacksonville': 1, 'Fort Worth': 1, 'Columbus': 1, 'Charlotte': 1, 'San Francisco': 1, 'Indianapolis': 1, 'Seattle': 1, 'Denver': 1, 'Washington': 1, 'Boston': 1, 'Nashville': 1, 'Baltimore': 1, 'Oklahoma City': 1, 'Las Vegas': 1, 'Portland': 1, 'Milwaukee': 1, 'Albuquerque': 1, 'Tucson': 1, 'Atlanta': 1})

4. Visit count by department:
defaultdict(<class 'int'>, {'Cardiology': 7, 'Orthopedics': 6, 'Dermatology': 5, 'Neurology': 5, 'General Medicine': 7})

5. Revenue by city:
[('Phoenix', 8280.0), ('Seattle', 7198.0), ('Jacksonville', 5824.0), ('Albuquerque', 3360.0), ('Las Vegas', 2990.0), ('Austin', 2645.0), ('Tucson', 2596.0), ('Fort Worth', 2006.0), ('San Francisco', 1725.0), ('Dallas', 1652.0), ('New York', 1416.0), ('Indianapolis', 1344.0), ('Portland', 1080.0), ('Charlotte', 1050.0), ('Washington', 1026.0), ('Los Angeles', 1008.0), ('Baltimore', 952.0), ('Milwaukee', 945.0), ('Atlanta', 892.5), ('Denver', 787.5), ('Chicago', 756.0), ('San Diego', 702.0), ('Nashville', 630.0), ('Philadelphia', 525.0), ('San Jose', 472.5)]

6. Revenue by department:
[('Neurology', 15640.0), ('Cardiology', 14868.0), ('Orthopedics', 12488.0), ('General Medicine', 5302.5), ('Dermatology', 3564.0)]

7. Top 3 paid visits:
[{'visit_id': 'V005', 'visit_date': '46206', 'patient_id': 'P105', 'patient_name': 'Robert Wilson', 'age': 62, 'city': 'Phoenix', 'department': 'Neurology', 'doctor_id': 'D204', 'visit_type': 'Emergency', 'diagnosis': 'Stroke Symptoms', 'bill_amount': 7200.0, 'payment_status': 'PAID', 'rating': 4, 'service_charge_pct': 0.15, 'service_charge': 1080.0, 'final_amount': 8280.0}, {'visit_id': 'V018', 'visit_date': '46212', 'patient_id': 'P118', 'patient_name': 'Rachel Robinson', 'age': 35, 'city': 'Seattle', 'department': 'Cardiology', 'doctor_id': 'D201', 'visit_type': 'Emergency', 'diagnosis': 'Shortness of Breath', 'bill_amount': 6100.0, 'payment_status': 'PAID', 'rating': 5, 'service_charge_pct': 0.18, 'service_charge': 1098.0, 'final_amount': 7198.0}, {'visit_id': 'V012', 'visit_date': '46209', 'patient_id': 'P112', 'patient_name': 'Megan White', 'age': 39, 'city': 'Jacksonville', 'department': 'Orthopedics', 'doctor_id': 'D202', 'visit_type': 'Emergency', 'diagnosis': 'Fracture', 'bill_amount': 5200.0, 'payment_status': 'PAID', 'rating': 4, 'service_charge_pct': 0.12, 'service_charge': 624.0, 'final_amount': 5824.0}]

8. Departments that handled emergency visits:
['Cardiology', 'Neurology', 'Orthopedics']
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9a. Distinct cities:
['Albuquerque', 'Atlanta', 'Austin', 'Baltimore', 'Boston', 'Charlotte', 'Chicago', 'Columbus', 'Dallas', 'Denver', 'Fort Worth', 'Houston', 'Indianapolis', 'Jacksonville', 'Las Vegas', 'Los Angeles', 'Milwaukee', 'Nashville', 'New York', 'Oklahoma City', 'Philadelphia', 'Phoenix', 'Portland', 'San Antonio', 'San Diego', 'San Francisco', 'San Jose', 'Seattle', 'Tucson', 'Washington']

9b. Distinct departments:
['Cardiology', 'Dermatology', 'General Medicine', 'Neurology', 'Orthopedics']

10. Department with highest average rating:
('General Medicine', 4.57)
```

Explanation for broadcast variable:

The broadcast variable stored the department service-charge percentages so Spark workers can efficiently access the same read-only lookup map. It is used to calculate each visit's service charge and final amount. The accumulators are updated during `foreach()` validation to count missing cities, invalid ages, invalid bill amounts, invalid payment statuses, and total invalid records.